Software Requirements Specification (SRS)

QR-Based Restaurant Menu Web Application

Project Information

Field	Details
Project Title	QR-Based Restaurant Menu Web Application
Course	SIT725 – Applied Software Engineering
Team Members:	Aroor Rohan Rao (Student ID: s226035073) Ferdinand Jacques Liauw (Student ID: s226255534) Avinash Shankaranarayanan (Student ID: s225596878) Ben Baiju (Student ID: s225709846)
Date	April 2026

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1. Introduction

1.1 Document Purpose

This Software Requirements Specification (SRS) document provides a detailed description of the requirements for the QR-Based Restaurant Menu Web Application. The purpose of this document is to define the functional and non-functional requirements of the system to guide developers, project managers, and stakeholders throughout the development lifecycle.

The intended audience includes:

- Developers and software engineers
- Project managers
- Academic evaluators
- System testers
- Restaurant stakeholders

1.2 Product Scope

The system is a web-based platform that allows restaurants in Melbourne to provide digital menus using QR codes. Customers can scan a QR code to access menus instantly.

The system includes:

- Customer-facing menu interface
- Admin dashboard for management
- Data-driven insights and recommendation features

Objectives:

- Enable contactless menu access
- Provide efficient restaurant management tools
- Deliver insights to improve business decisions

1.3 Document Overview

This document includes:

1. Introduction
2. Overall Description
3. Specific Requirements
4. Supporting Information

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
QR Code	A machine-readable code used to access the menu
Admin	Authorized user managing restaurants and menus
Customer	End user viewing the menu
MERN Stack	MongoDB, Express, React, Node.js
API	Application Programming Interface
UI	User Interface
UX	User Experience
CRUD	Create, Read, Update, Delete
AI	Artificial Intelligence

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2. Overall Description

2.1 Product Perspective

The system is a standalone web application that integrates with restaurant operations. It replaces traditional paper menus with a digital solution accessible via QR codes.

It operates as:

- A centralized platform managing multiple restaurants
- A cloud-based system accessible via browsers
- A modular system with separate frontend and backend

2.2 Product Functions

The system will provide the following high-level features:

- QR code generation for each restaurant
- Digital menu display for customers
- Admin dashboard for restaurant and menu management
- Categorization of menu items (e.g., starters, mains, drinks)
- Image and description support for menu items
- AI-based recommendations (e.g., popular items)
- Basic analytics (e.g., most viewed items)

2.3 User Characteristics

Admin (System Administrator)

- Has technical knowledge
- Manages platform-level operations
- Can add and manage multiple restaurants
- Responsible for maintaining system data

Customers

- General public users
- No technical expertise required
- Access the system via smartphones
- Interact only with the menu interface

Restaurant Owners

- Non-technical or semi-technical user
- Uses the system to manage their own restaurant
- Can add, update, and delete menu items
- Focused on business insights rather than system configuration
- Uses the analytics dashboard to:
    - Monitor popular dishes
    - Track category performance
    - Analyze customer interaction trends

2.4 Constraints

- Must use MERN stack
- No external paid APIs
- Mobile-first design
- Web-based only
- Limited development timeframe

2.5 Assumptions and Dependencies

Assumptions:

- Internet access available
- Users can scan QR codes

Dependencies:

- MongoDB database
- Hosting platform
- QR code generation library

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3. Specific Requirements

3.1 External Interfaces

User Interfaces

- Responsive web UI
- Admin dashboard
- Customer menu interface

Hardware Interfaces

- Smartphones for scanning QR codes

Software Interfaces

- MongoDB
- Node.js APIs
- QR code library

3.2 Functional Requirements

3.2.1 Admin Functional Requirements

- Admin shall be able to log in securely
- Admin shall be able to create and manage restaurant accounts
- Admin shall be able to assign access to restaurant owners
- Admin shall be able to generate QR codes for restaurants
- Admin shall be able to manage platform-level data
- Admin shall be able to monitor overall system usage

3.2.2 Restaurant Owner Functional Requirements

- Restaurant owner shall be able to log in securely
- Restaurant owner shall be able to manage their restaurant profile
- Restaurant owner shall be able to add, update, and delete menu items
- Restaurant owner shall be able to categorize menu items
- Restaurant owner shall be able to upload images and descriptions
- Restaurant owner shall be able to view their restaurant's QR code

3.2.3 Analytics & Insights Requirements

- Restaurant owner shall be able to view analytics for their restaurant
- Restaurant owner shall be able to view most viewed menu items
- Restaurant owner shall be able to view category-wise performance
- Restaurant owner shall be able to analyze customer interaction trends
- Restaurant owner shall be able to view usage patterns over time
- Restaurant owner shall be able to identify popular items

3.2.4 Customer Functional Requirements

- Customer shall be able to scan QR codes
- Customer shall be redirected to the menu page
- Customer shall be able to view categorized menu items
- Customer shall be able to view item details
- Customer shall be able to search and filter items
- Customer shall be able to make orders.

3.2.5 AI / Intelligent Features

- System shall highlight popular items dynamically
- System shall provide recommendation sections (e.g., “Trending Items”)
- System shall support basic automated description generation

3.3 Non-Functional Requirements

3.3.1 Performance

- Menu loads within 2–3 seconds
- Analytics data loads efficiently without delay
- Supports concurrent users

3.3.2 Security

- Admin authentication required
- Secure API endpoints

3.3.3 Usability

- Mobile-friendly UI
- Simple navigation

3.3.4 Reliability

- 95% uptime
- Error handling

3.3.5 Scalability

- Supports multiple restaurants
- Expandable database

3.3.6 Maintainability

- Modular architecture
- Clean code separation

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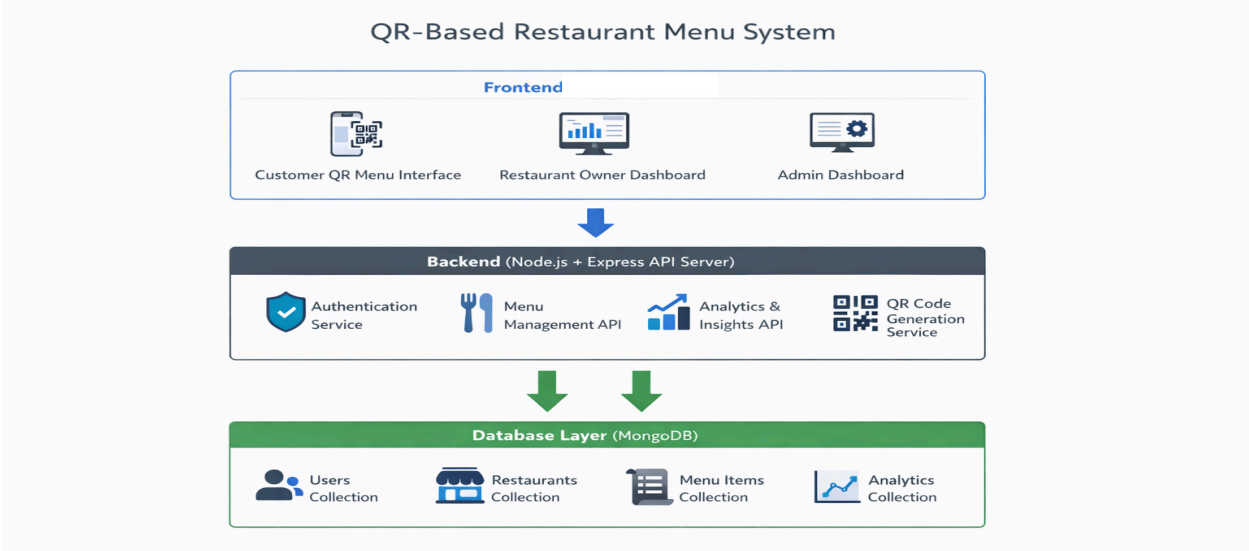
4. Supporting Information

4.1 System Architecture Diagram

The system follows a simple web-based client-server architecture, separating the frontend, backend, and database for clarity and maintainability.

System Components:

- Frontend (HTML, CSS, Materialize): QR-based menu interface for customers and basic admin views for menu management.
- Backend (Node.js + Express.js): REST APIs, authentication, analytics, QR code generation
- Database (MongoDB): Stores restaurant data, menu items, users, and analytics



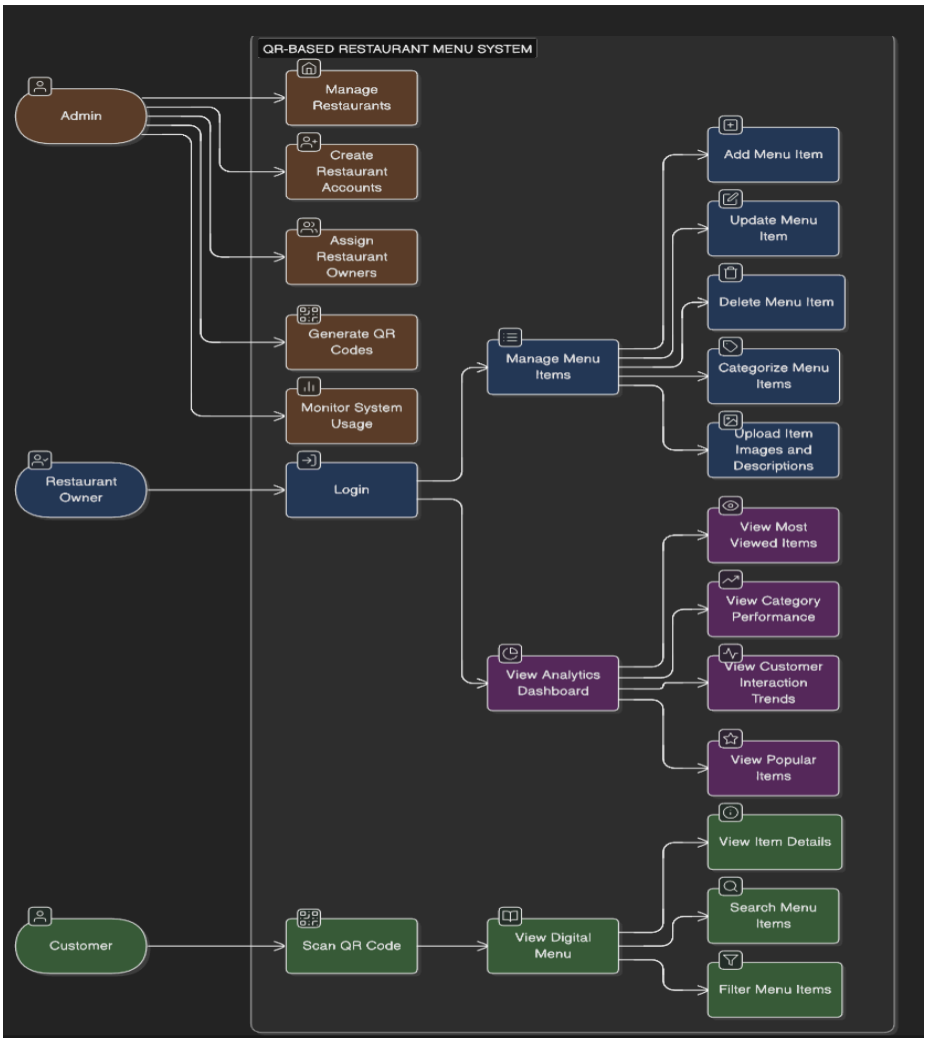
4.2 Use Case Diagram

The system has three main actors:

- Admin
- Restaurant Owner
- Customer

Key interactions include:

- Admin managing restaurants and assigning owners
- Restaurant Owner managing menu and viewing analytics
- Customer scanning QR code and viewing menu



4.3 API Documentation

The system follows a **RESTful API architecture** to enable communication between the frontend (React) and backend (Node.js + Express.js). These APIs handle authentication, restaurant management, menu operations, analytics, and QR code generation.

Authentication APIs

Used for secure login and access control for Admin and Restaurant Owners.

- **POST /api/auth/register**
  Registers a new user (Admin or Restaurant Owner)
- **POST /api/auth/login**
  Authenticates user and returns a JWT token
- **GET /api/auth/profile**
  Retrieves logged-in user details

Restaurant Management APIs

Used for managing restaurant data.

- **POST /api/restaurants**
  Create a new restaurant (Admin only)
- **GET /api/restaurants/:id**
  Retrieve restaurant details
- **PUT /api/restaurants/:id**
  Update restaurant information
- **DELETE /api/restaurants/:id**
  Delete a restaurant

Menu Management APIs

Used by Restaurant Owners to manage menu items.

- **POST /api/menu/:restaurantId**
  Add a new menu item
- **GET /api/menu/:restaurantId**
  Retrieve all menu items for a restaurant
- **PUT /api/menu/item/:itemId**
  Update a menu item

- **DELETE /api/menu/item/:itemId**
  Delete a menu item

Analytics APIs

Used to fetch insights for Restaurant Owners.

- **GET /api/analytics/:restaurantId**
  Retrieve analytics dashboard data
- **GET /api/analytics/popular-items/:restaurantId**
  Get most viewed/popular items
- **GET /api/analytics/trends/:restaurantId**
  Get customer interaction trends over time

QR Code API

Used to generate QR codes for restaurants.

- **POST /api/qrcode/generate/:restaurantId**
  Generates a QR code linking to the restaurant menu

API Security

- JWT-based authentication is used
- Role-based access control (Admin / Owner / Customer)
- Protected routes for sensitive operations
- Input validation to prevent injection attacks

4.4 Legal and Ethical Compliance

Data Privacy

- The system does not collect unnecessary personal customer data
- Any collected data (e.g., analytics) is stored in aggregated form
- No personally identifiable information (PII) is shared or sold

Security Compliance

- Secure authentication using encrypted JWT tokens
- Passwords are hashed using industry-standard hashing algorithms
- Role-based access control prevents unauthorized access

Ethical Considerations

- Analytics are used only to improve restaurant decision-making
- Customer behaviour tracking is anonymised
- No invasive tracking or third-party data sharing is used

Intellectual Property

- Restaurants retain ownership of their menu content and branding
- The platform acts only as a service provider for digital menu hosting

Accessibility Considerations

- Web interface designed to be mobile-friendly
- Simple UI to support users with low technical experience
- QR-based access removes need for app installation

4.5 References

- Greenleaf, G, Waters, N, Lane, K, Arnold, B and Clarke, R 2020, 'Bringing Australia's Privacy Act up to international standards (submission in response to the Privacy Act review - issues paper)', *SSRN Electronic Journal*, viewed 11 April 2026, https://doi.org/10.2139/ssrn.3752152
- Iskender, A, Sirakaya-Turk, E and Cardenas, D 2023, 'Restaurant menus and COVID-19: implications for technology adoption in the post-pandemic era', *Consumer Behavior in Tourism and Hospitality*, vol. 18, no. 4, pp. 587–605, viewed 11 April 2026, https://doi.org/10.1108/CBTH-11-2022-0194
- Node.js Documentation https://nodejs.org
- Express.js Documentation https://expressjs.com

- MongoDB Documentation https://www.mongodb.com
- React Documentation https://react.dev
- JSON Web Token (JWT) Documentation https://jwt.io
- QR Code Generator Libraries (npm ecosystem) https://www.npmjs.com
- REST API Design Principles https://restfulapi.net
- Software Engineering Lecture Materials (SIT725)

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5. Team Members Contributions

Ben Baiju (s225709846)

- Project ideation and initial concept development
- Product Scope and Document Overview
- Assumptions and Dependencies
- API Documentation and endpoint design

Aroor Rohan Rao (s226035073)

- Product Perspective and Product Functions
- User Characteristics
- Functional Requirements (all user types)
- Use Case Diagram development and review

Ferdinand Jacques Liauw (s226255534)

- Non-Functional Requirements
- External Interfaces
- System Architecture Diagram development and review

Avinash Shankaranarayanan (s225596878)

- Legal and Ethical Compliance
- Academic reference sourcing and APA 7 formatting
- Member Contributions section and document formatting and consolidation